Data Analyst

Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

Dataset Overview

Here are the key characteristics of the dataset:

- **Shape**: 167,826 rows × 9 columns
- **Columns**:
  - `Country/Region`: Name of the country or region
  - `UNICEF_Region`: Regional classification
  - `Indicator`: Description of the HIV-related metric
  - `Year`: The year the data corresponds to
  - `Sex`: Gender (`M`, `F`, `Both`)
  - `Age`: Age group (`0-14`, `15-19`, etc.)
  - `Value`, `Lower`, `Upper`: Estimates of the indicator value and its confidence interval (many are represented as strings like `<0.01`)
- **Missing Data**:
  - `UNICEF_Region` has ~10k missing values
  - `Lower` and `Upper` confidence bounds have 57 missing entries
- **Data Types**:
  - `Year` is numeric (`float64`), while `Value`, `Lower`, and `Upper` are strings, potentially requiring conversion for analysis.

Let me know the specific analytical question you'd like to explore using this dataset. [>_]

Which country has the lowest HIV prevalence among those in the highest quartile for new infections?

It looks like the filtering step returned an empty set when trying to find the country with the lowest HIV prevalence among those in the highest quartile for new infections. This may be due to inconsistencies in how "incidence" and "prevalence" indicators are labeled or perhaps missing/invalid numeric values after parsing.

Let me fix the filtering logic to ensure we're capturing the correct indicators and re-run the analysis. [>_]

Based on the indicator names, we should refine our filters as follows:

- **Incidence**: `"Estimated number of annual new HIV infections"`
- **Prevalence**: `"Estimated number of people living with HIV"`

I'll re-run the analysis using these exact indicator labels. [>_]

Answer: Among countries in the highest quartile for new HIV infections, **Lesotho** has the **lowest number of people living with HIV**, with an estimated **645 individuals** (in the latest available data year).

Python Code Used

python Copiar Editar

```python
# Apply precise filters for incidence and prevalence
incidence_data = df_latest[df_latest['Indicator'] == 'Estimated number of annual new HIV 
prevalence_data = df_latest[df_latest['Indicator'] == 'Estimated number of people living 

# Group by country and calculate mean values
incidence_avg = incidence_data.groupby('Country/Region')['Value_clean'].mean()
prevalence_avg = prevalence_data.groupby('Country/Region')['Value_clean'].mean()

# Identify high incidence countries (top 25%)
incidence_threshold = incidence_avg.quantile(0.75)
high_incidence_countries = incidence_avg[incidence_avg >= incidence_threshold].index

# Among those, find the one with the lowest prevalence
relevant_prevalence = prevalence_avg[prevalence_avg.index.isin(high_incidence_countries)]
lowest_prevalence_country = relevant_prevalence.idxmin()
lowest_prevalence_value = relevant_prevalence.min()
```

Let me know if you'd like to visualize this or explore another angle. [>_]

Envie uma mensagem para o Data Analyst